Expected value calculations

ADVANCED PROBABILITY: UNCERTAINTY IN DATA



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What is expected value?

- Weighted average
- Expectation over time
- Take into account multiple possibilities



How expected value calculations work

1. Identify possible outcomes
2. Assign probabilities
3. Calculate the expected value
4. Interpret the results and make a decision

- A lottery ticket: win (USD 1,000) or lose the ticket price (USD 2)
- Winning: 0.1%, losing: 99.9%
- Expected value:

$$(0.001 * 1000) + (0.999 * 2) = -1$$

- On average, you will lose money if you keep buying lottery tickets!

Expected value calculations in practice

- Bigger picture: possibilities and their probabilities
- Balance risk and reward
- Example applications:
  - Set product prices
  - Assess financial impact of different strategies
  - Compare profitability of stock options



How are probabilities for expected values determined?

- Historical data analysis:
  - Past trends provide probabilities
- Market research and surveys:
  - Market data provides probabilities
- Expert judgment:
  - Expert estimations provide probabilities



Example: pricing decisions

Demand	Units Sold	Probability	Revenue ($)
High	1000	30%	50,000
Moderate	700	50%	35,000
Low	400	20%	20,000
Total Expected Revenue			**36,500**

- Company can:
  - Compare expected revenue to break-even point
  - Reconsider pricing strategy

Let's practice!

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Interpreting confidence and prediction intervals

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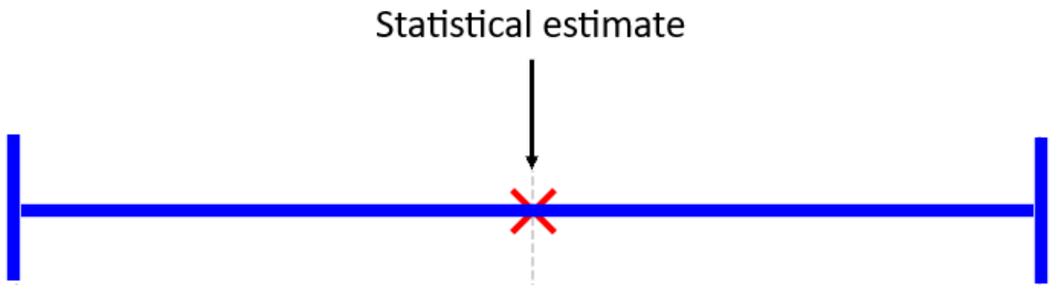
Why do we need intervals?

- Data-driven decisions involve uncertainty
- Variability in the data and outcomes
- Confidence and prediction intervals help communicate uncertainty

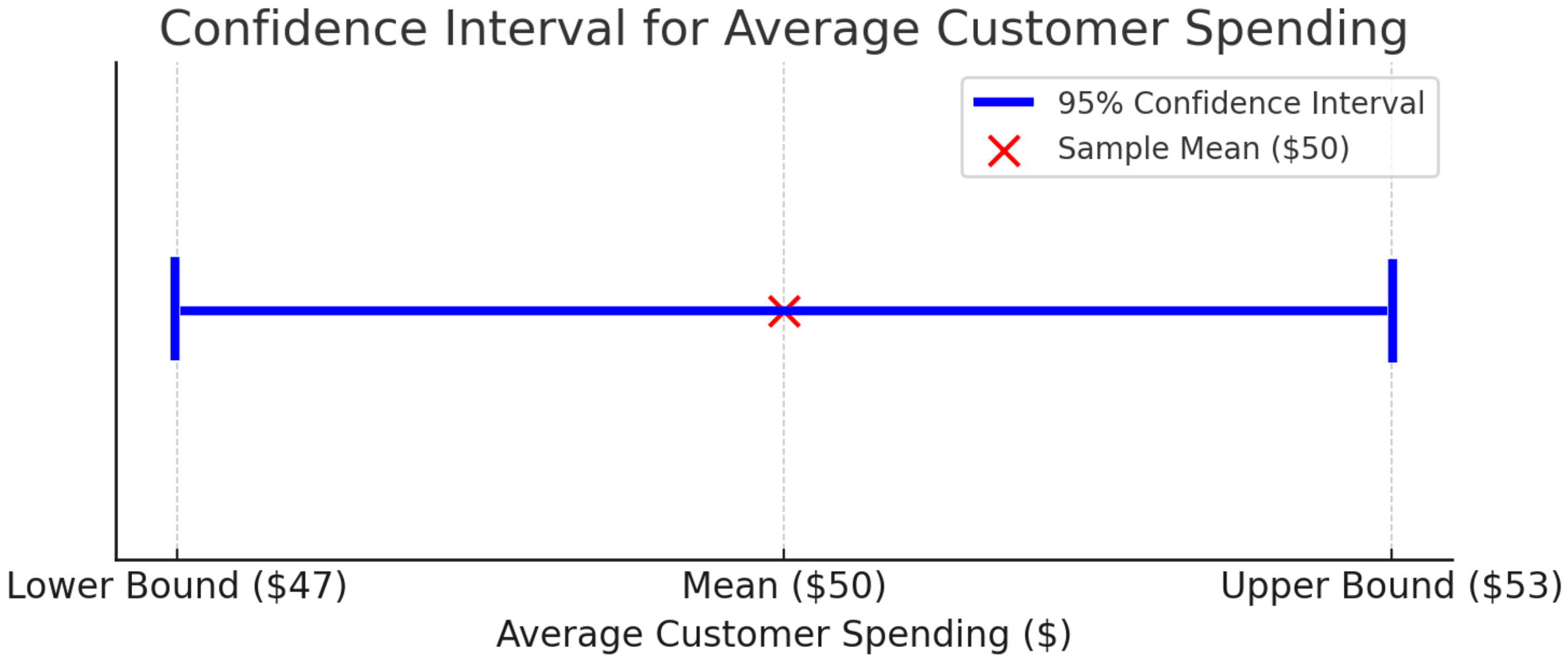


Confidence intervals

- Reliability of estimated statistical parameter
- Based on sampling process

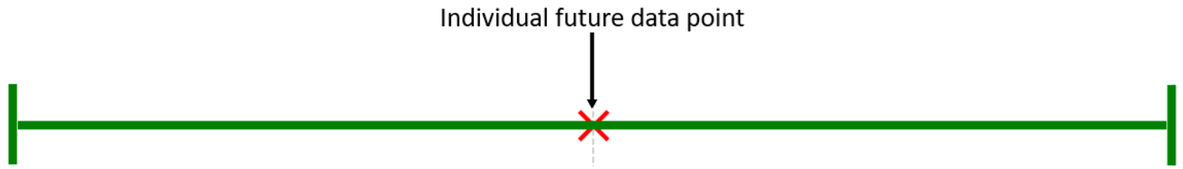


Confidence interval example

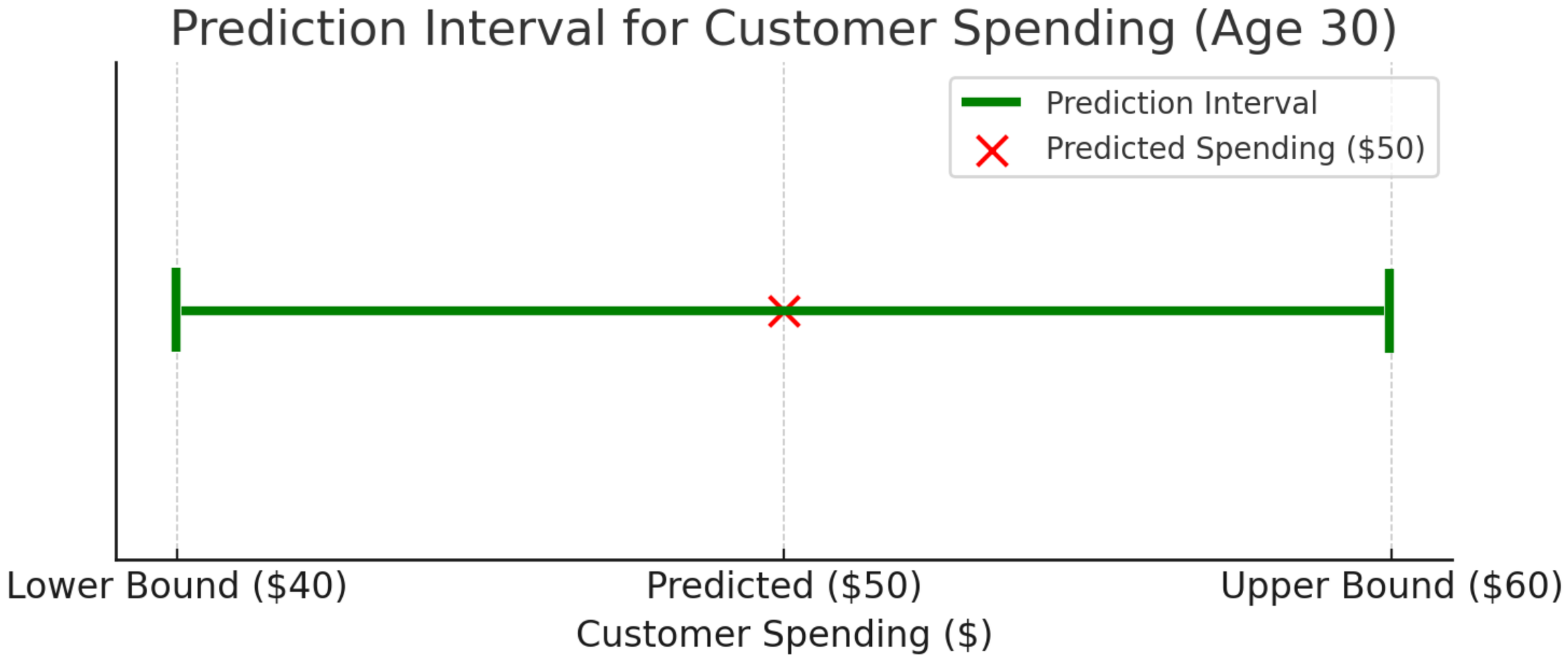


Prediction intervals

- Range for individual new data point
- Based on statistical model
- Typically wider



Prediction interval example



Confidence vs. prediction intervals

Confidence interval

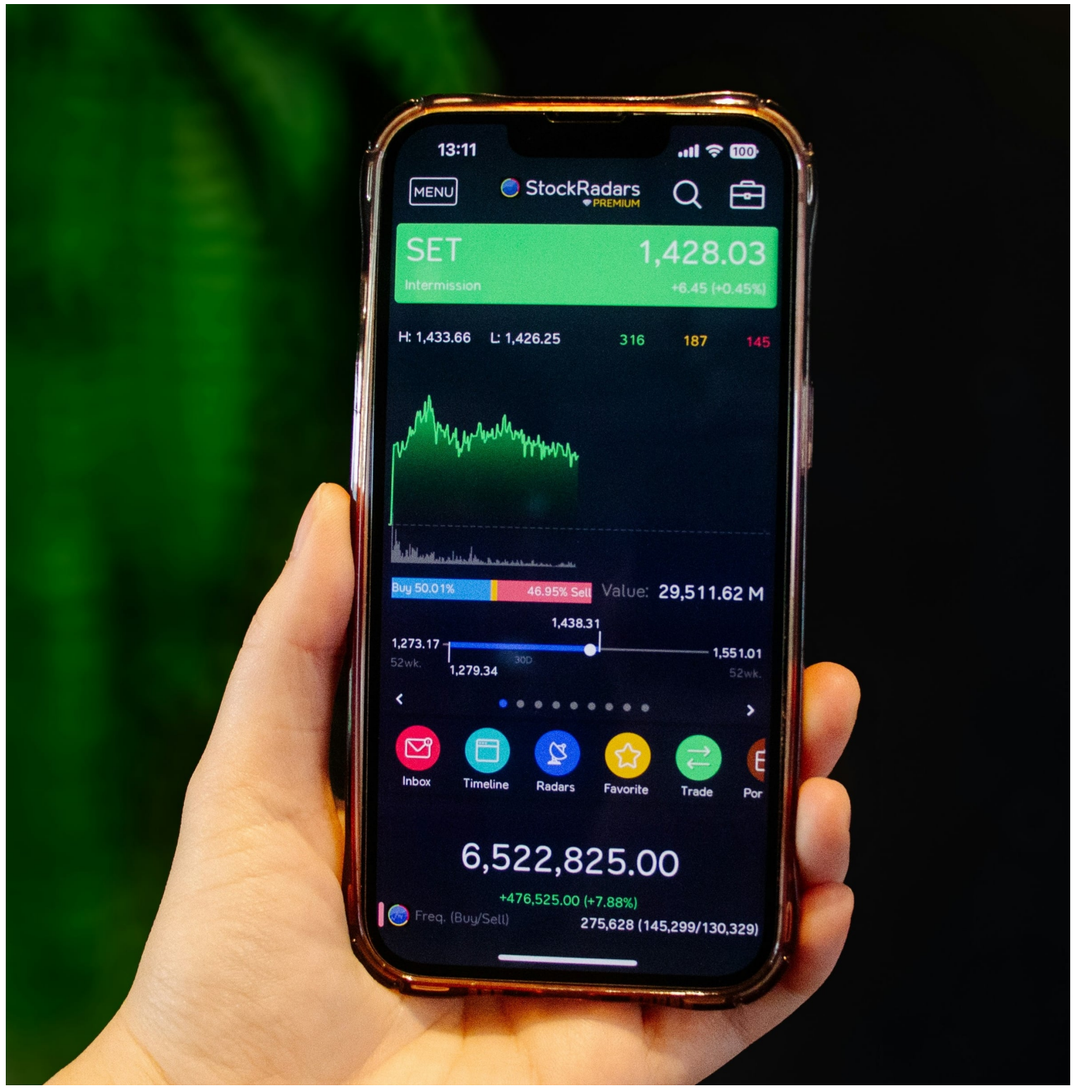
- Range for a statistical parameter
- Based on sampling process
- Typically narrower

Prediction interval

- Range for an individual observation
- Based on a predictive model
- Typically wider

Example: risk management

- Return on investment portfolio
- Average annual return of 8%



Example: risk management

Interval Type	Purpose	95% Interval Range
Confidence Interval	Portfolio's average return estimate	7% to 9%
Prediction Interval	Single investment return estimate	4% to 12%

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Scenario analysis

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What is scenario analysis?

- Strategic decision-making method
- How do different combinations of conditions affect outcomes?
- Prepare for multiple future scenarios



Scenario types

- **Baseline scenario:** things stay the same
- **Best-case scenario:** optimistic view
- **Worst-case scenario:** pessimistic view



Applications of scenario analysis

- What happens if...?
- How will this change if this changes?
- What can we expect best-case or worst-case?



- Examples:
  - What happens if inflation rises faster than expected?
  - How will our profits change if demand decreases by 20%?
  - What's our best and worst-case revenue

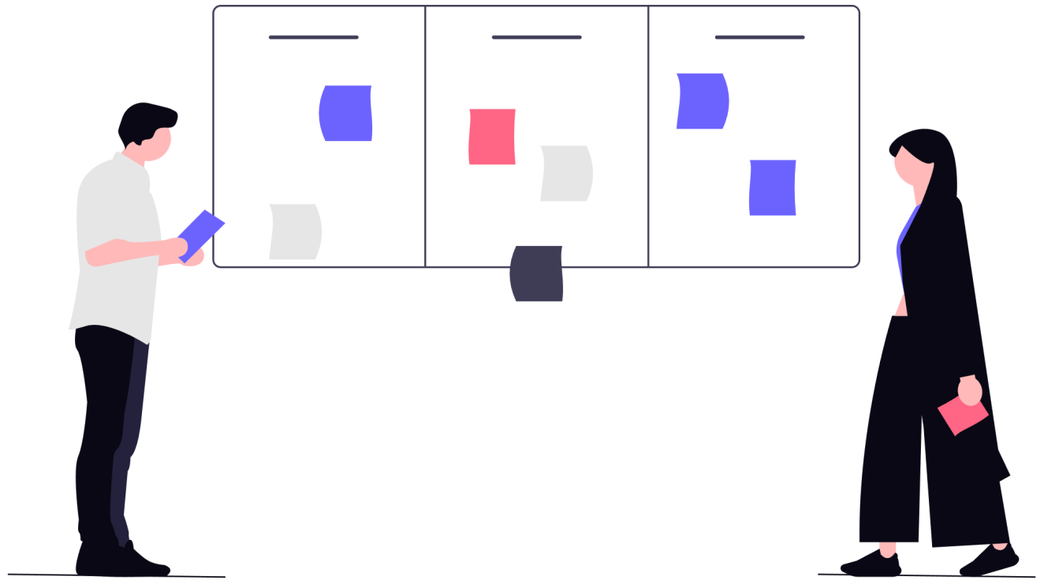
How to conduct scenario analysis

1. Identify key variables



How to conduct scenario analysis

1. Identify key variables
2. Define possible scenario's



How to conduct scenario analysis

1. Identify key variables
2. Define possible scenario's
3. Model the outcomes



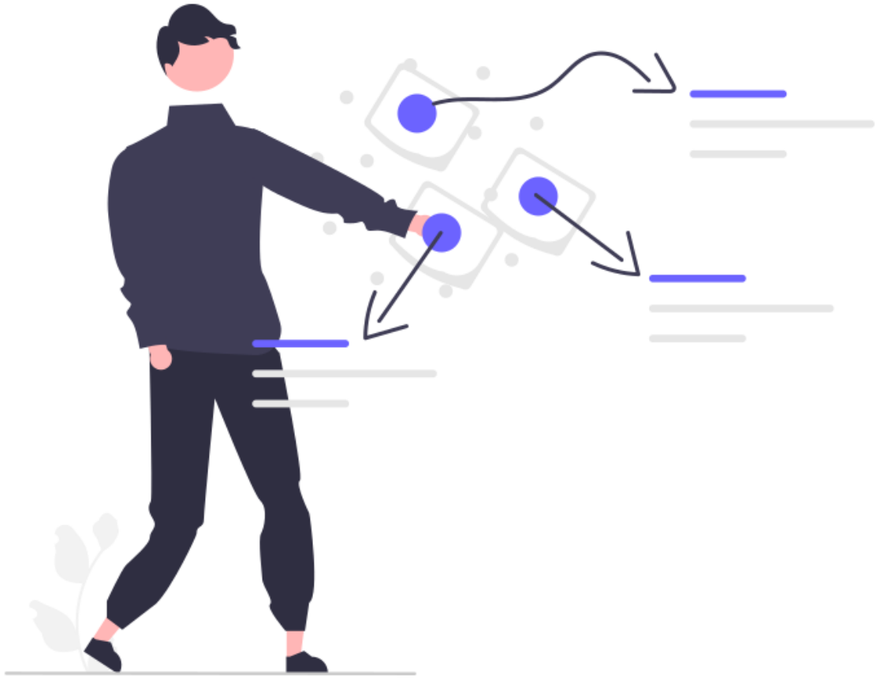
How to conduct scenario analysis

1. Identify key variables
2. Define possible scenario's
3. Model the outcomes
4. Analyze and compare results



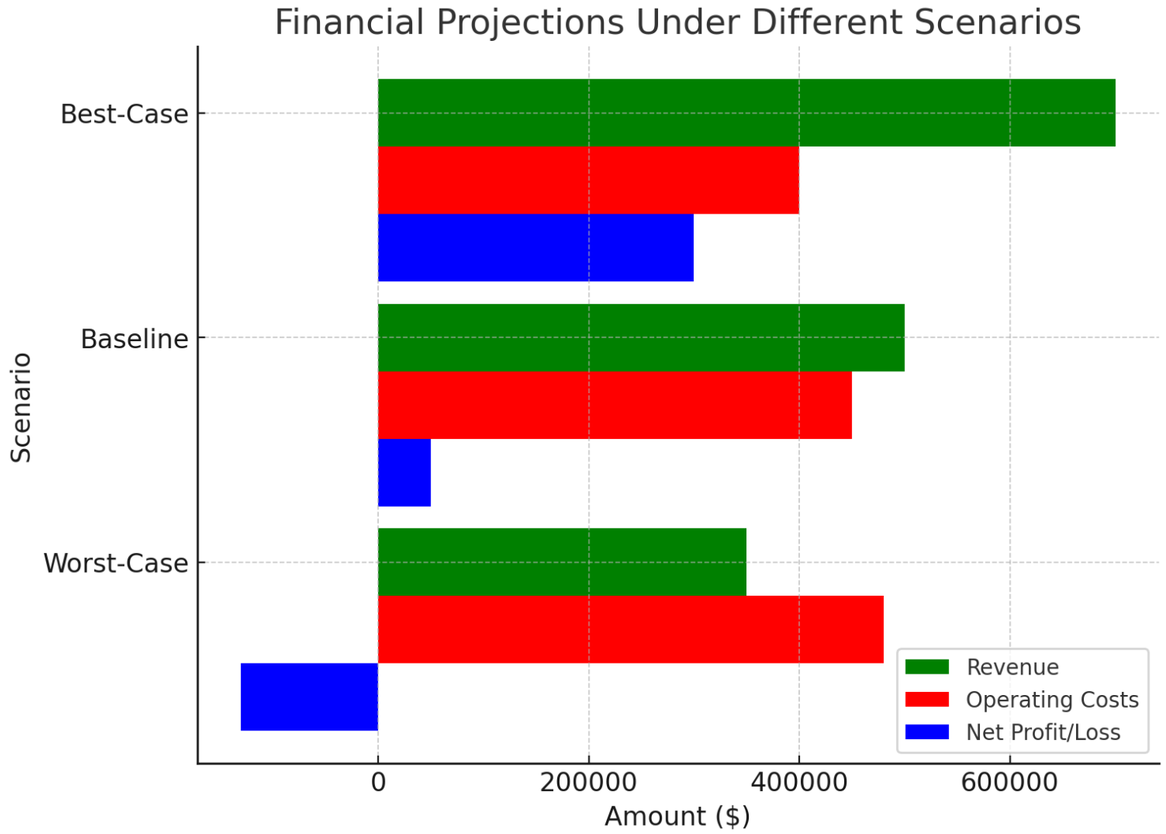
How to conduct scenario analysis

1. Identify key variables
2. Define possible scenario's
3. Model the outcomes
4. Analyze and compare results
5. Develop strategic responses



Example: Retail business revenue forecast

Scenario	Projected Monthly Revenue	Projected Operating Costs	Projected Net Profit/Loss
Best-Case	$700,000	$400,000	+$300,000
Baseline	$500,000	$450,000	+$50,000
Worst-Case	$350,000	$480,000	-$130,000



Example: Retail business revenue forecast

The retail company plans accordingly:

- **High demand:** increase marketing efforts, expand inventory, open more stores
- **Stable demand:** proceed as usual and monitor
- **Low demand:** cut costs, essential products only, delay expansion



Let's practice!

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Sensitivity analysis

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What is sensitivity analysis?

- How does one variable change...
- ... while keeping the others constant?
- Goal: identify key drivers
- Enhance pro-active decision-making



Sensitivity analysis vs. scenario analysis

Scenario analysis

- Multiple variables at once
- Explores multiple possible future outcomes
- Best-case, baseline, and worst-case scenarios

Sensitivity analysis

- Focus on one specific variable
- Measures how changes in one variable affect the outcome
- Range of possible outcomes due to variable changes

Steps of sensitivity analysis

1. Identify key variables



Steps of sensitivity analysis

1. Identify key variables
2. Establish reference scenario



Steps of sensitivity analysis

1. Identify key variables
2. Establish reference scenario
3. Adjust one variable at a time

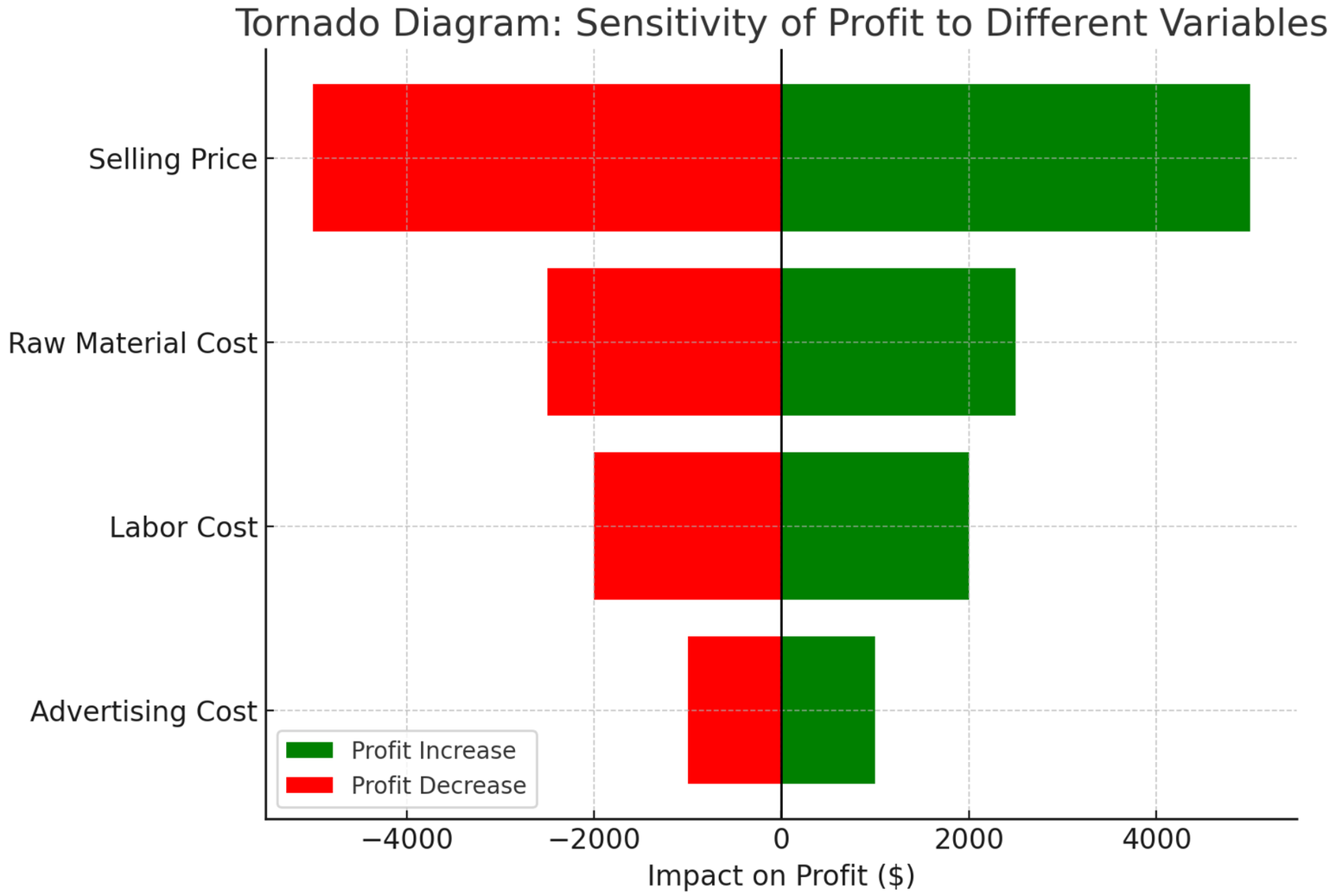


Steps of sensitivity analysis

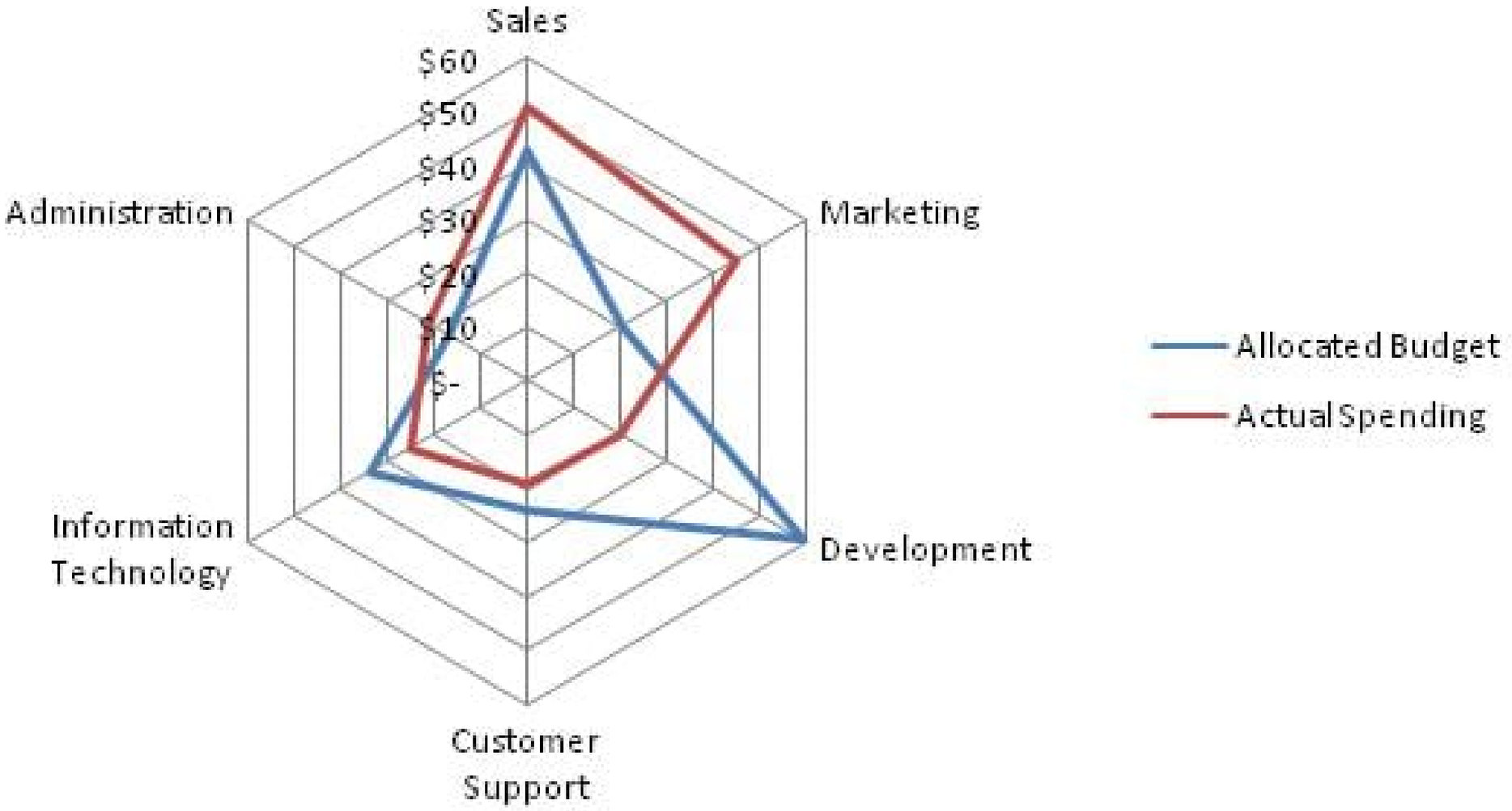
1. Identify key variables
2. Establish reference scenario
3. Adjust one variable at a time
4. Analyze the results



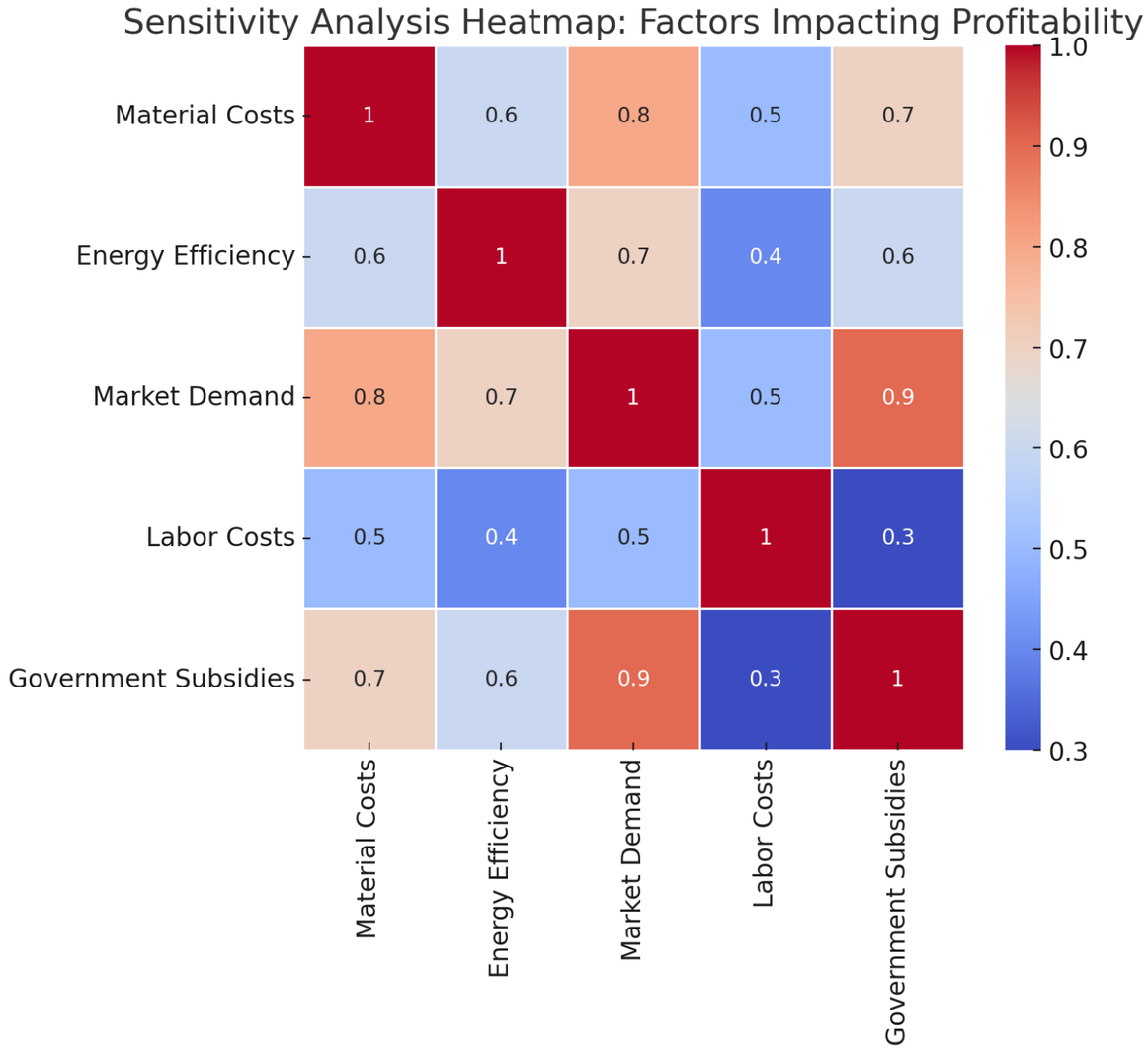
Visualizing the results of sensitivity analysis



Visualizing the results of sensitivity analysis



Visualizing the results of sensitivity analysis

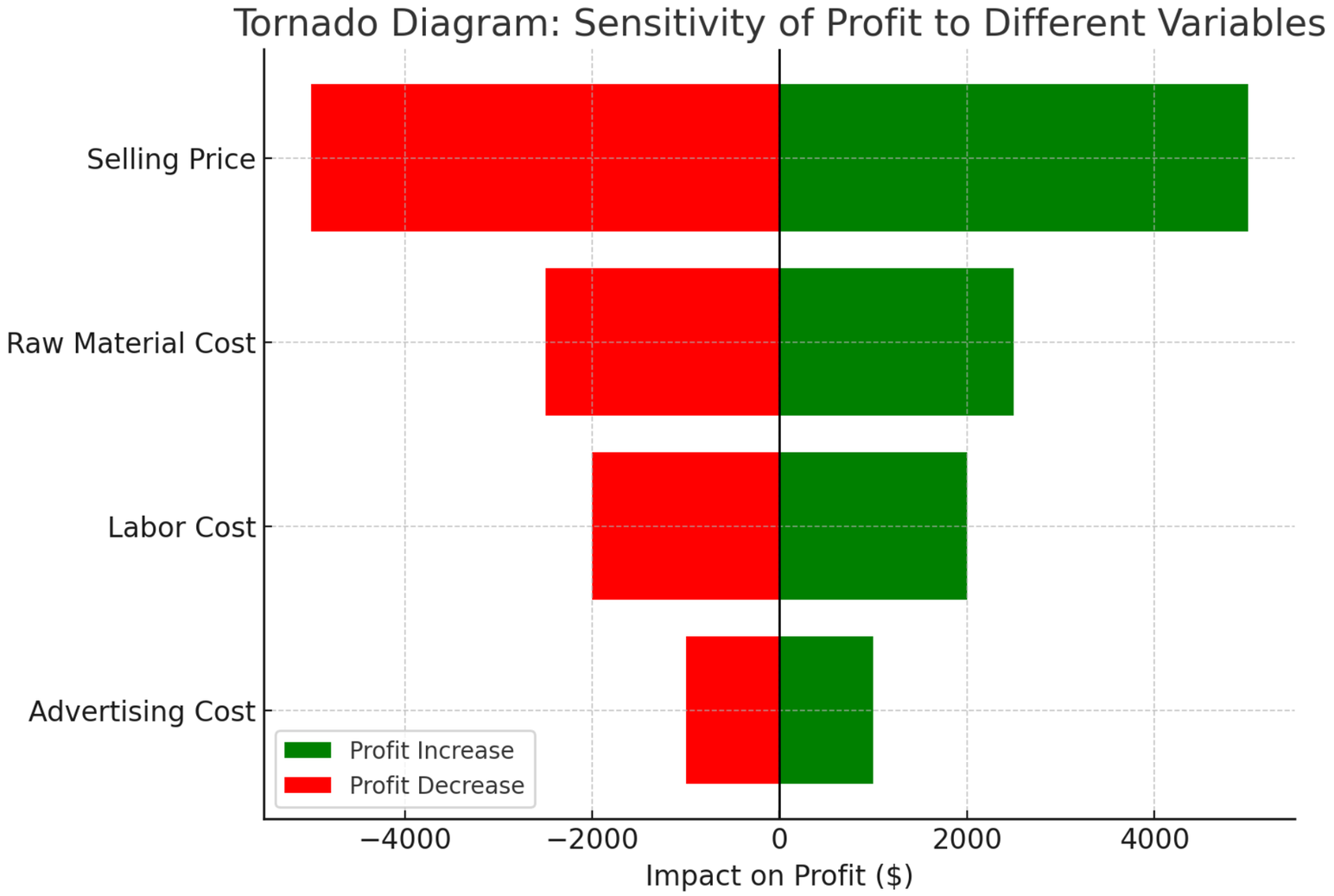


Example: Profit margins

- How do different cost and revenue factors impact profit margins?
- Sensitivity analysis to test changes in:
  - Product price
  - Raw material cost
  - Labor cost
  - Advertising spend



Example: Profit margins



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